

Time-varying Minimum-Variance Portfolios and Dynamic CVaR Extensions

A complete mathematical, computational, empirical, and statistical account

Independent reproduction-and-extension project

6 September 2026

Abstract

This report explains the entire repository as a self-contained research project. It first reconstructs the mathematical and computational logic of Fan, Wu, Yang, and Zhong’s time-varying minimum-variance portfolio (TV-MVP), using the authors’ public manual and unchanged MATLAB/R release. It then documents the portable Python reproduction, a dynamic factor-covariance extension, empirical factor-bootstrap CVaR, a constraint-matched comparison, and a filtered, stressed, regularised CVaR repair. Every estimation date uses only trailing information. The empirical study uses 39 long-history stocks selected from a current NIFTY 50 list, a 504-trading-day window, and monthly rebalancing. Statistical uncertainty is evaluated with paired moving-block bootstrap intervals on monthly losses; regimes, covariance forecasts, factor diagnostics, tail calibration, simulations, and parameter sensitivity are also studied. The principal empirical conclusion is deliberately limited: raw empirical CVaR is poorly calibrated and unstable after 2021; filtered scenarios and regularisation materially improve its point estimates, but do not establish statistically reliable superiority over a constraint-matched MVP. Equal weight has the highest realised return and Sharpe ratio in this sample. The project is a rigorous exploratory extension, not evidence of future profitability.

Interpretation note. The words “original”, “benchmark”, and “author-style” refer to the public methodology and code release. The journal paper’s original proprietary empirical dataset is not in the repository. Consequently, the project reproduces the public simulation/example workflow as closely as portable software allows; it does not claim an exact replication of every journal table.

Contents

1	Executive overview	6
1.1	The scientific question	6
1.2	The strategies	6
1.3	Main numerical findings	6
2	Repository provenance, organisation, and research scope	7
2.1	What was supplied	7
2.2	Independent project structure	7
2.3	Claims that are and are not in scope	8
3	Data, timing, and information sets	8
3.1	Author example data	8

3.2	NIFTY empirical snapshot	8
3.3	Rolling chronology and no-look-ahead rule	9
4	Base-paper mathematical framework	9
4.1	Locally time-varying factor model	9
4.2	Boundary-adjusted Epanechnikov weighting	9
4.3	Local principal components	10
4.4	Selecting the factor number	10
4.5	Residual covariance estimation	10
4.6	Factor covariance and the TV-MVP covariance	11
4.7	Minimum-variance weights	11
4.8	Author diagnostic tests	11
5	Portable reproduction	12
5.1	Simulation equations	12
5.2	Reproduction results and interpretation	12
6	Extension 1: dynamic factor covariance	13
6.1	Motivation	13
6.2	EWMA estimator	13
7	Extension 2: scenario-based CVaR	14
7.1	Loss, VaR, and expected shortfall	14
7.2	Baseline scenario generator	14
7.3	Finite-scenario linear program	14
7.4	Why CVaR need not maximise return	14
8	Filtered, stressed, and regularised CVaR	15
8.1	Filtered historical innovations	15
8.2	Regularised CVaR objective	15
8.3	Positive-definite repair	15
9	Backtest accounting and performance measures	16
9.1	Holdings drift and turnover	16
9.2	Return and risk definitions	16
10	Statistical analysis	16
10.1	Monthly paired moving-block bootstrap	16
10.2	Ex-ante volatility regimes	17

10.3 Covariance forecast quality	17
10.4 Factor diagnostics	18
11 Empirical results	18
11.1 Core full-sample comparison	18
11.2 Primary monthly inference	19
11.3 Regime results	19
11.4 Covariance forecast evaluation	20
11.5 Factor-model diagnostics	20
12 Why empirical CVaR failed after 2021	21
12.1 Removing the feasible-set confound	21
12.2 Forecast-tail reversal	21
12.3 Corner solutions and costs	22
13 Improved CVaR results	22
13.1 Ablation results	22
13.2 Calibration after the repair	23
13.3 Statistical uncertainty	23
13.4 High- versus normal-volatility performance	23
14 Simulation evidence	24
14.1 Design	24
14.2 Results	25
15 Robustness and sensitivity	26
16 Software implementation in detail	26
16.1 Module-to-method map	26
16.2 Numerical solvers	27
16.3 Randomness and reproducibility	28
16.4 Tests	28
17 Limitations, threats to validity, and honest interpretation	28
17.1 Data validity	28
17.2 Model validity	28
17.3 Evaluation validity	29
17.4 Economic interpretation	29
18 Research programme for a publishable extension	29

19 Reproduction instructions	30
A Algorithmic pseudocode	30
A.1 Common rolling backtest	30
A.2 Moving-block inference	31
B Complete configuration snapshot	31
C Result provenance	32
D Glossary	33

List of Tables

1	Strategy definitions	6
2	Top-level project organisation	7
3	Portable reproduction versus manual targets	12
4	Core NIFTY results, February 2017–August 2026	18
5	Pre-specified core comparisons using monthly losses	19
6	Mean next-holding-period covariance forecast losses	20
7	Matched post-2021 comparison	21
8	Post-2021 improved-CVaR ablation; all optimisers matched	22
9	Post-2021 tail calibration and concentration	23
10	Probability of beating original TV-MVP over 100 replications	25
11	Python implementation map	26
12	Key configuration values	31

List of Figures

1	Reproduction diagnostics. Loading trajectories are visibly time-varying, while portable thresholding recovers broad residual-covariance structure but not the author’s exact penalised estimator.	13
2	Full-sample cumulative net wealth for the core comparison.	19
3	Rolling 63-day annualised volatility and empirical ES. The common temporal movement dominates the modest cross-strategy differences.	20
4	Factor diagnostics. Large principal angles and heavy-tailed scores/residuals are directly relevant to covariance and CVaR instability.	21
5	Cumulative net wealth for the matched-constraint experiment, over the full period and after 2021.	22

6	Post-2021 improved-CVaR diagnostics. The shared legend identifies all six series. Regularised filtered CVaR improves wealth and turnover-related stability, but risk paths remain close and calibration gaps still spike in some months.	24
7	Mean realised variance and ES ratios to original TV-MVP. Values below one favour the displayed method, subject to the constraint caveat.	25
8	Compact robustness heatmap. It shows dynamic-minus-original Sharpe only; CVaR sensitivity remains available in the CSV table.	26

1 Executive overview

1.1 The scientific question

For a portfolio of p assets, the minimum-variance problem depends on the conditional return covariance matrix. A conventional sample covariance assumes that the joint distribution is sufficiently stable inside the estimation window. Fan et al. relax that assumption by allowing factor loadings to vary smoothly over time while keeping factor and residual covariance components constant over the local estimation period [1]. Their estimator is

$$\hat{\Sigma}_{r,t}^{\text{TV}} = \hat{B}_t \hat{\Sigma}_F \hat{B}_t^{\text{T}} + \hat{\Sigma}_e. \quad (1)$$

The first repository extension lets factor risk itself change:

$$\hat{\Sigma}_{r,t}^{\text{dyn}} = \hat{B}_t \hat{\Sigma}_{F,t}^{\text{EWMA}} \hat{B}_t^{\text{T}} + \hat{\Sigma}_e. \quad (2)$$

The second extension stops treating variance as the only risk criterion. It creates return scenarios from the fitted local factor model and chooses weights to minimise 95% loss CVaR. A final follow-up addresses observed scenario-tail miscalibration with filtered historical simulation, recency weighting, explicit stress draws, and L_1 portfolio regularisation.

1.2 The strategies

The repository uses the following labels.

Table 1: Strategy definitions

Label	Construction	Default feasible set
A: Equal weight	$w_i = 1/p$ at each rebalance	Long only; no active cap issue because $1/39 < 10\%$
B: Original TV-MVP	Equation (1), then minimum variance	Unconstrained in the core run
C: Dynamic- Σ_F TV-MVP	Equation (2), then minimum variance	Unconstrained in the core run
D: Empirical CVaR	Paired block-bootstrap factor/residual scenarios	Long only; 10% maximum weight
D-FS: Filtered/stressed CVaR	Current-state rescaling, recent blocks, stress mixture	Long only; 10% cap
D-REG: Regularised filtered CVaR	D-FS plus turnover and diversification penalties	Long only; 10% cap

The core A–D comparison intentionally preserves the requested original unconstrained MVP benchmark, but it is not a pure objective comparison because D has a different feasible set. The later matched experiment gives B, C, and all CVaR variants identical long-only, 10%-cap constraints. Results from the matched experiment are the appropriate answer to “MVP versus CVaR under the same portfolio rules.”

1.3 Main numerical findings

Across the full February 2017–August 2026 out-of-sample period, equal weight has the highest annualised return and Sharpe ratio. Dynamic factor covariance is extremely close to the original

covariance estimator. Raw empirical CVaR has slightly lower full-sample daily volatility and ES than the unconstrained benchmark, but lower return and deeper drawdown.

After 2021, the fair matched comparison reverses the apparent tail-risk advantage: empirical CVaR has higher realised volatility, ES, drawdown, and turnover than both matched MVPs. Its fitted scenario CVaR is lower, but subsequent realised tail loss is higher. This is evidence of scenario-tail estimation error, not a linear-programming failure.

The pre-specified repair raises empirical CVaR’s post-2021 annualised net return from 10.17% to 14.30% and lowers annual turnover from 4.75 to 3.50. Its fitted-versus-realised daily CVaR gap falls from 0.219 percentage point to 0.005 percentage point. Nevertheless, its paired monthly differences from matched original MVP all have 95% bootstrap intervals containing zero. The correct conclusion is “promising exploratory repair”, not “superior strategy.”

2 Repository provenance, organisation, and research scope

2.1 What was supplied

The initial repository was an author code release comprising 26 files: a 16-page user manual, MATLAB functions, one R bridge script, a Windows batch file, two MATLAB data files, and an example CSV. Every supplied file was moved without content changes into `original_author_code/`. Byte-level SHA-256 comparisons against Git history were performed for all 26 files after reorganisation.

The author workflow assumes MATLAB, CVX, Windows `dos`, and R packages `spcov`, `PDSCE`, and `R.matlab`. The extension is implemented in Python because that dependency chain is not portable in the present environment. The Python implementation is not passed off as the literal author executable.

2.2 Independent project structure

Table 2: Top-level project organisation

Path	Purpose
<code>original_author_code/</code>	Untouched author MATLAB/R release and manual
<code>src/tvmvp_ext/</code>	Portable estimation, optimisation, backtesting, inference, diagnostics, and simulation library
<code>scripts/</code>	Reproduction, empirical, diagnostic, matched, improved-CVaR, simulation, and all-in-one entry points
<code>data/raw/</code>	Frozen adjusted-close and NIFTY index snapshots
<code>results/tables/</code>	Machine-readable numerical evidence
<code>results/plots/</code>	Saved diagnostics and performance figures
<code>tests/</code>	Constraint, timing, covariance, bootstrap, and reproducibility tests
<code>config.yaml</code>	Single source of model and evaluation parameters
<code>01_...md-06_...md</code>	Concise staged reports and follow-up investigations
<code>report/</code>	This complete LaTeX source

2.3 Claims that are and are not in scope

The project can support claims about the behaviour of these implementations on the frozen sample and stated DGPs. It cannot establish that any portfolio will outperform in the future, that results survive point-in-time universe construction, or that the portable sparse estimator is numerically identical to the author R estimator. The improved CVaR specification was fixed in `config.yaml` before its one reported run, but it was designed after observing the earlier NIFTY failure. It is therefore internally pre-specified for the follow-up comparison but not an externally registered, fully independent confirmatory test.

3 Data, timing, and information sets

3.1 Author example data

The supplied `example_data.mat` contains a 50×500 return matrix named `R0`. The manual's example uses observations 1–250 for the first estimate, holds for five days, and repeats weekly through observation 500. It reports daily Sharpe-like ratios, means, and standard deviations for TV-MVP and $1/N$.

The synthetic author DGP uses $p = 50$, $T = 200$ or 250 , two AR(1) factors, smooth stochastic loadings, and Toeplitz residual covariance $(\Sigma_e)_{ij} = 0.5^{|i-j|}$. Section 5 gives the exact equations and reproduction outcome.

3.2 NIFTY empirical snapshot

The extension requested adjusted-close returns beginning in 2015. The frozen data manifest records:

- source: Yahoo Finance adjusted-close field retrieved through `yfinance`;
- requested dates: 1 January 2015 through 5 September 2026, with the end date exclusive at download;
- observed prices: 1 January 2015 through 4 September 2026;
- requested universe: 40 names drawn from a current NIFTY 50 constituent list;
- retained universe: 39 names; `TRENT.NS` returned no usable history;
- minimum completeness rule: 98.5%; all retained series were recorded as complete before limited filling;
- frozen stock-file SHA-256: `e6d40f9f3eba...8a55`;
- frozen index-file SHA-256: `b0f0307ea507...571e`.

Adjusted-close simple returns are

$$r_{i,t} = \frac{P_{i,t}^{\text{adj}}}{P_{i,t-1}^{\text{adj}}} - 1. \quad (3)$$

Prices and the NIFTY index are aligned on common dates. At most three consecutive missing prices are forward-filled, then any row with a remaining stock missing is removed. Returns are not converted to excess returns because no daily risk-free series is present. Consequently, “Sharpe ratio” in this project is mean return divided by standard deviation with a zero risk-free rate, not the exact excess-return Sharpe ratio used in some finance studies.

Important limitation. The stock universe is selected using current constituents and long histories. Delisted firms and past constituents that later left the index are absent. This creates survivorship and lookback-selection bias even though the rolling estimators themselves do not use future returns. A publishable confirmatory study should use point-in-time membership and delisting returns.

3.3 Rolling chronology and no-look-ahead rule

Let d_j be the last trading date of month j . With window length $W = 504$, the estimator at d_j receives only

$$\mathcal{I}_{d_j} = \{r_{d_j-W+1}, \dots, r_{d_j}\}. \quad (4)$$

Weights are computed after d_j and take effect on the next trading date. They remain in force, with natural within-period drift, through d_{j+1} . No return dated after d_j enters estimation or scenario generation for that rebalance. A dedicated test perturbs the last ten observations and verifies that all earlier weights remain identical.

The out-of-sample sample contains 2,362 trading days, from 1 February 2017 through 31 August 2026, generated by 115 rebalance decisions dated 31 January 2017 through 31 July 2026. September 2026 is excluded because a complete next monthly holding interval was not available.

4 Base-paper mathematical framework

4.1 Locally time-varying factor model

Let $r_t = (r_{1t}, \dots, r_{pt})^\top \in \mathbb{R}^p$ denote returns and $f_t \in \mathbb{R}^K$ latent common factors. The base model is

$$r_t = B_t f_t + e_t, \quad B_t = B(t/T), \quad (5)$$

where $B(\cdot)$ changes smoothly in rescaled time, while e_t contains idiosyncratic returns. A convenient identification is $T^{-1} \sum_t f_t f_t^\top = I_K$, with factor rotations otherwise observationally equivalent. Under contemporaneous factor-residual orthogonality,

$$\Sigma_{r,t} = \text{Cov}(r_t | \mathcal{I}_{t-1}) = B_t \Sigma_F B_t^\top + \Sigma_e, \quad (6)$$

The authors' key modelling choice is that time variation in $\Sigma_{r,t}$ is transmitted primarily through B_t , whereas Σ_F and Σ_e are estimated over the window as constant. This is more flexible than a constant-loading factor model, but less flexible than a fully time-varying covariance decomposition.

4.2 Boundary-adjusted Epanechnikov weighting

For target date r and observation t , define $u_{tr} = (t - r)/(Th)$. The interior Epanechnikov kernel is

$$K(u) = \frac{3}{4}(1 - u^2)\mathbf{1}(|u| \leq 1). \quad (7)$$

The author `Kernel2.m` divides the kernel by its truncated integral near the left and right sample boundaries, following the boundary-correction idea cited to Hong and Li and Su and Wang [5, 4]. In schematic form,

$$K_r^*(u) = \begin{cases} K(u) / \int_{-r/(Th)}^1 K(v) dv, & r < Th, \\ K(u), & Th \leq r \leq T - Th, \\ K(u) / \int_{-1}^{(1-r/T)/h} K(v) dv, & r > T - Th. \end{cases} \quad (8)$$

The MATLAB weight is $k_{h,tr}^* = h^{-1}K_r^*(u_{tr})$. The Python port uses the same compact-support kernel but normalises the discrete weights to sum to one. Because factor scores and loadings can be reciprocally rescaled, the resulting common covariance is the meaningful object for comparison.

The rule-of-thumb bandwidth is

$$h = \frac{2.35}{\sqrt{12}} T^{-1/5} p^{-1/10}. \quad (9)$$

4.3 Local principal components

Store returns as $R = [r_1, \dots, r_T] \in \mathbb{R}^{p \times T}$. At target r , form the weighted data matrix

$$X_r = D_r R^\top, \quad D_r = \text{diag}(\sqrt{k_{h,1r}^*}, \dots, \sqrt{k_{h,T_r}^*}). \quad (10)$$

Local PCA solves the rank- K approximation problem

$$\min_{F_r, \Lambda_r} \|X_r - F_r \Lambda_r\|_F^2, \quad \text{s.t.} \quad T^{-1} F_r^\top F_r = I_K. \quad (11)$$

If $U_{r,K}$ contains the first K eigenvectors of $X_r X_r^\top$, then the author code uses

$$\hat{F}_r = \sqrt{T} U_{r,K}, \quad \hat{\Lambda}_r = T^{-1} \hat{F}_r^\top X_r. \quad (12)$$

The $p \times K$ loading matrix is $\hat{B}_r = \hat{\Lambda}_r^\top$. PCA eigenvector signs are arbitrary. The author code applies a sequential sign convention; the Python port aligns each factor's loading vector to the previous target by its inner-product sign. Sign alignment changes plots, not $B \Sigma_F B^\top$.

After estimating the complete loading path, contemporaneous factor scores are least-squares projections:

$$\hat{f}_t = (\hat{B}_t^\top \hat{B}_t)^{-1} \hat{B}_t^\top r_t, \quad \hat{e}_t = r_t - \hat{B}_t \hat{f}_t. \quad (13)$$

The implementation uses a stable least-squares solver rather than an explicit matrix inverse.

4.4 Selecting the factor number

The manual gives a BIC-type criterion

$$\hat{K} = \arg \min_{1 \leq m \leq K_{\max}} \left\{ \log V(m) + m \frac{p+Th}{pTh} \log \left(\frac{pTh}{p+Th} \right) \right\}. \quad (14)$$

The author implementation computes a specialised loss V using a nested CVX optimisation. The portable selector replaces V with mean squared local-PCA residual loss while retaining the penalty form. This is a documented approximation, not an exact numerical reimplementation. The empirical study fixes $K = 3$ in the configuration and treats $K \in \{2, 3, 5\}$ as robustness analysis. The simulation reproduction searches up to six factors and selects two.

4.5 Residual covariance estimation

Let

$$S_e = \frac{1}{T-1} \sum_{t=1}^T (\hat{e}_t - \bar{e})(\hat{e}_t - \bar{e})^\top. \quad (15)$$

The author workflow first obtains a positive-definite starting value with `PDSCE::pdsoft`, then calls `spcov`. The underlying sparse-covariance approach approximately minimises a penalised Gaussian covariance objective of the form

$$\log \det \Sigma + \text{tr}(S_e \Sigma^{-1}) + \lambda \sum_{i \neq j} |\sigma_{ij}|, \quad \Sigma \succeq \delta I, \quad (16)$$

using positive-definite algorithms described by Bien and Tibshirani and Rothman [7, 8]. The diagonal is not penalised. The author selects λ by three-fold likelihood cross-validation.

The portable estimator instead thresholds residual correlations. With $D_e = \text{diag}(s_1, \dots, s_p)$ and $C = D_e^{-1} S_e D_e^{-1}$,

$$c_0 = a \sqrt{\frac{\log p}{T}}, \quad a = 0.15, \quad (17)$$

$$\tilde{C}_{ij} = \text{sign}(C_{ij}) \max(|C_{ij}| - c_0, 0), \quad i \neq j, \quad (18)$$

$$\tilde{C}_{ii} = 1, \quad \tilde{\Sigma}_e = D_e \tilde{C} D_e. \quad (19)$$

This is simpler and portable but not equal to (16). Positive definiteness is imposed afterward as described in Section 8.3.

4.6 Factor covariance and the TV-MVP covariance

The original factor covariance estimate is the ordinary sample covariance of the score vectors:

$$\hat{\Sigma}_F = \frac{1}{T-1} \sum_{t=1}^T (\hat{f}_t - \bar{f})(\hat{f}_t - \bar{f})^\top. \quad (20)$$

At the endpoint, the estimator is Equation (1). All past score observations affect $\hat{\Sigma}_F$ equally; only $\hat{B}_T$ is explicitly local in time.

4.7 Minimum-variance weights

The fully invested minimum-variance problem is

$$\min_w w^\top \hat{\Sigma} w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1. \quad (21)$$

For an invertible covariance matrix and no other constraints, the closed-form solution is

$$w_{\text{MVP}} = \frac{\hat{\Sigma}^{-1} \mathbf{1}}{\mathbf{1}^\top \hat{\Sigma}^{-1} \mathbf{1}}. \quad (22)$$

The author MATLAB function solves (21) through CVX. The Python code solves the linear system for the unconstrained case. For the matched experiment it solves the quadratic program with

$$\mathbf{1}^\top w = 1, \quad 0 \leq w_i \leq 0.10. \quad (23)$$

4.8 Author diagnostic tests

The release contains a two-step structural testing procedure. Step 1 compares local-factor fitted values with a constant-loading PCA fit. In the code,

$$\widehat{M} = \frac{1}{pT} \sum_{i=1}^p \sum_{t=1}^T \left(\widehat{b}_{i,t}^\top \hat{f}_t - \widehat{b}_i^\top \tilde{f}_t \right)^2, \quad (24)$$

and a centred, scaled statistic J_{pT} uses estimated bias and variance terms. Its finite-sample reference distribution is generated by 199 Gaussian-residual bootstrap samples under constant loadings. Step 2 splits residuals into two halves, compares the first two spectral moments of

their sample covariance matrices, and evaluates a quadratic form against a χ^2_2 critical value. The manual applies a conservative split of the overall significance level between steps.

These hypothesis tests are preserved in the author folder but are not ported into the empirical extension. The extension instead reports loading-space angles and residual-correlation diagnostics. Therefore, diagnostic evidence of instability in this report is not a reproduction of the authors' formal two-step test.

5 Portable reproduction

5.1 Simulation equations

The author DGP has two loading functions. For $u = t/T$ and asset i ,

$$b_{i,t}^{(1)} = \rho[(3 + z_{i1})u + z_{i2} \sin(4\pi u)] + z_{i3}, \quad (25)$$

$$b_{i,t}^{(2)} = \rho[(3 + v_{i1})u + v_{i2} \sin(4\pi u) + (v_{i3}u)^2] + v_{i4}, \quad (26)$$

with independent standard-normal coefficients. Factors follow

$$f_{1,t} = 0.6f_{1,t-1} + \epsilon_{1,t}, \quad \epsilon_{1,t} \sim N(0, 0.64), \quad (27)$$

$$f_{2,t} = 0.3f_{2,t-1} + \epsilon_{2,t}, \quad \epsilon_{2,t} \sim N(0, 0.91), \quad (28)$$

after a 200-observation burn-in. Both stationary factor variances equal one. Residuals are Gaussian with $(\Sigma_e)_{ij} = 0.5^{|i-j|}$, and

$$r_t = b_t^{(1)} f_{1,t} + b_t^{(2)} f_{2,t} + e_t. \quad (29)$$

The Python DGP preserves these equations but uses NumPy's random-number generator and project seed 20260906, not MATLAB's generator with seed 2. This intentionally makes the run reproducible inside the new project, but it prevents exact equality with the manual's simulated numeric values.

5.2 Reproduction results and interpretation

Table 3: Portable reproduction versus manual targets

Metric	Portable	Manual	Difference
Selected factor number	2	2	0
Oracle forecast variance	0.2270	0.1373	0.0897
TV-MVP forecast variance	0.2426	0.1000	0.1426
Sample-covariance forecast variance	0.3190	0.2117	0.1073
TV-MVP weight mean absolute error	0.0431	—	—
Sample weight mean absolute error	0.0758	—	—
Residual covariance relative Frobenius error	0.4824	—	—
Weekly TV-MVP daily Sharpe-like ratio	0.0378	0.0865	-0.0487
Weekly equal-weight daily Sharpe-like ratio	0.0495	0.0488	0.0007

The qualitative result that local TV-MVP weights are closer to oracle weights than sample-covariance weights is reproduced: mean absolute error is 0.0431 versus 0.0758. The selected

factor count also matches. The absolute simulated variances do not match, principally because the RNG stream and sparse covariance algorithm differ. On the supplied real-data example, equal weight closely matches the manual’s mean and Sharpe-like ratio, but portable TV-MVP does not reproduce the manual’s claimed advantage. This is reported as a difference, not hidden.

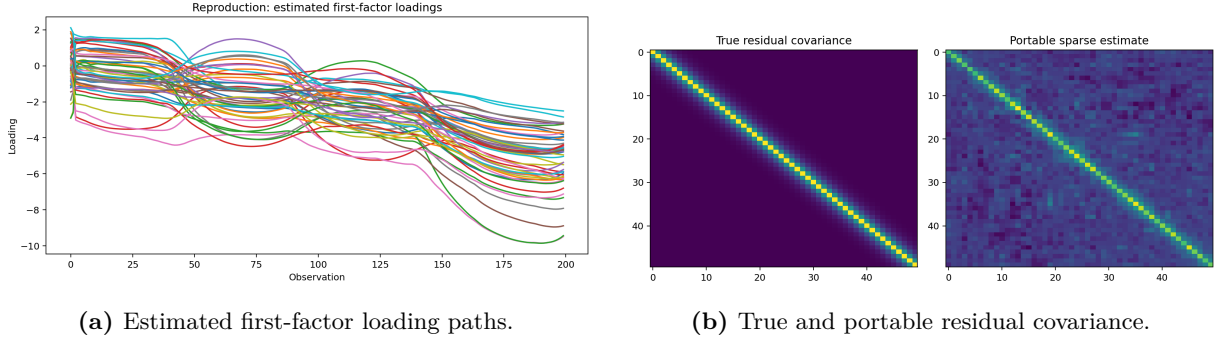


Figure 1: Reproduction diagnostics. Loading trajectories are visibly time-varying, while portable thresholding recovers broad residual-covariance structure but not the author’s exact penalised estimator.

6 Extension 1: dynamic factor covariance

6.1 Motivation

Equation (1) treats the factor covariance as constant across the complete rolling window. If the volatility of common shocks rises or falls faster than the loading functions, equal weighting of historical scores can lag. The minimal extension changes only the factor covariance term, leaving local loadings and sparse residual covariance untouched. This makes C a clean diagnostic of whether recency-weighted factor risk adds value beyond time-varying exposures.

6.2 EWMA estimator

For decay $\lambda \in (0, 1)$ and T estimated score vectors, define

$$a_t = \frac{\lambda^{T-t}}{\sum_{j=1}^T \lambda^{T-j}}, \quad \bar{f}_a = \sum_{t=1}^T a_t \hat{f}_t. \quad (30)$$

The weighted covariance includes the finite-weight degrees correction

$$\hat{\Sigma}_{F,T}^{\text{EWMA}} = \frac{\sum_{t=1}^T a_t (\hat{f}_t - \bar{f}_a)(\hat{f}_t - \bar{f}_a)^\top}{1 - \sum_{t=1}^T a_t^2}. \quad (31)$$

The effective number of observations is

$$T_{\text{eff}} = \left(\sum_t a_t^2 \right)^{-1}. \quad (32)$$

With $\lambda = 0.94$ and $T = 504$, $T_{\text{eff}} = 32.33$. Thus the extension reacts quickly but estimates factor covariance from an amount of information comparable to only 32 equally weighted observations. This variance-bias trade-off is central to interpreting the weak empirical improvement.

7 Extension 2: scenario-based CVaR

7.1 Loss, VaR, and expected shortfall

For portfolio return $R_p = w^\top r$, define loss $L = -R_p$. At confidence β ,

$$\text{VaR}_\beta(L) = \inf\{\ell : \Pr(L \leq \ell) \geq \beta\}. \quad (33)$$

For continuous losses, expected shortfall is

$$\text{ES}_\beta(L) = \mathbb{E}[L \mid L \geq \text{VaR}_\beta(L)]. \quad (34)$$

Expected shortfall is a coherent tail-risk measure under standard definitions [11]. Rockafellar and Uryasev express CVaR through the convex function [9, 10]

$$\Phi_\beta(w, \alpha) = \alpha + \frac{1}{1 - \beta} \mathbb{E}[(L(w) - \alpha)_+]. \quad (35)$$

Minimising (35) jointly over w and α minimises CVaR; α acts as a VaR threshold.

7.2 Baseline scenario generator

At each rebalance, only the trailing fitted scores $\{\hat{f}_t\}$, residual vectors $\{\hat{e}_t\}$, and current endpoint loading $\hat{B}_T$ are available. A circular moving-block bootstrap chooses a uniform start index and appends $L = 10$ consecutive paired observations, wrapping at the sample end, until $S = 600$ scenario dates are obtained. Scores and residual vectors use the same indices, preserving their contemporaneous empirical relationship and short-run time ordering. Scenario s is

$$r^{(s)} = \hat{B}_T \hat{f}_{J_s} + \hat{e}_{J_s}. \quad (36)$$

This generator is empirical: it can rearrange historical blocks but cannot create a shock magnitude absent from the 504-day window. It also maps all past factors through today's loading while leaving historical score and residual scales unfiltered.

7.3 Finite-scenario linear program

For scenario returns $r^{(s)}$ and $\beta = 0.95$, the baseline optimisation is

$$\min_{w, \alpha, u} \quad \alpha + \frac{1}{(1 - \beta)S} \sum_{s=1}^S u_s, \quad (37)$$

$$\text{s.t.} \quad u_s \geq -r^{(s)\top} w - \alpha, \quad u_s \geq 0, \quad (38)$$

$$\mathbf{1}^\top w = 1, \quad 0 \leq w_i \leq 0.10. \quad (39)$$

It is solved with SciPy HiGHS. At 95% and 600 scenarios, only roughly 30 observations lie in the nominal tail. Block dependence makes the effective amount of independent tail information smaller still. An optimiser can exploit accidental differences among these few tail observations, producing unstable corner portfolios even when the LP is solved exactly.

7.4 Why CVaR need not maximise return

Program (37) contains no expected-return term. A stock that avoided losses in the fitted tail can receive weight even if it has weak subsequent upside. Conversely, a stock with high expected return but several historical losses is penalised. CVaR is not “wrong” when such a portfolio lags a rally; it is optimising a different functional. The meaningful failure occurs when it also fails on realised tail loss, which is what the matched post-2021 experiment finds.

8 Filtered, stressed, and regularised CVaR

8.1 Filtered historical innovations

Let S_f and S_e be the unweighted sample covariances of centred fitted scores and residuals. Symmetric whitening produces

$$z_t^f = (\hat{f}_t - \bar{f})S_f^{-1/2}, \quad z_t^e = (\hat{e}_t - \bar{e})S_e^{-1/2}. \quad (40)$$

Small eigenvalues are floored before inverse square roots. Current target factor covariance is Equation (31). Current residual covariance is

$$Q_{e,T} = \omega \hat{\Sigma}_{e,T}^{\text{EWMA}} + (1 - \omega) \hat{\Sigma}_e^{\text{sparse}}, \quad \omega = 0.50, \quad (41)$$

where residual EWMA decay is 0.97. The mixture is repaired to be positive definite.

Circular block starts now have probability

$$\pi_j = \frac{\delta^{T-1-j}}{\sum_{q=0}^{T-1} \delta^q}, \quad \delta = 0.995, \quad (42)$$

so recent blocks are more likely. Pairing remains intact. With $S = 800$ and stress indicator $I_s \sim \text{Bernoulli}(0.15)$, the filtered scenario is

$$r^{(s)} = \hat{B}_T \left[m_f(I_s) z_{J_s}^f Q_{f,T}^{1/2} \right] + m_e(I_s) z_{J_s}^e Q_{e,T}^{1/2}, \quad (43)$$

where $m_f(0) = m_e(0) = 1$, $m_f(1) = 1.50$, and $m_e(1) = 1.15$. Because common-factor shocks receive the larger multiplier, stressed draws increase the common component's share of total variance and thereby tend to raise cross-asset correlation as well as volatility. This is a transparent stress mixture, not an estimated regime-transition probability.

8.2 Regularised CVaR objective

The final variant solves

$$\min_{w, \alpha, u} \widehat{\text{CVaR}}_{0.95}(w) + \eta_{\text{turn}} \|w - w^-\|_1 + \eta_{\text{div}} \left\| w - \frac{1}{p} \mathbf{1} \right\|_1, \quad (44)$$

subject to Equation (23), with $\eta_{\text{turn}} = 0.0010$ and $\eta_{\text{div}} = 0.00025$. Here w^- is the naturally drifted pre-trade portfolio. Auxiliary nonnegative variables linearise both absolute-value penalties, so the entire problem remains a linear program.

The first penalty makes small forecast changes insufficient to trigger large trades. The second shrinks toward equal weight and discourages needless zeros and cap hits. It is an L_1 distance penalty, not a direct penalty on the Herfindahl index; “diversification regularisation” should therefore be read as shrinkage toward $1/p$.

8.3 Positive-definite repair

Every covariance is symmetrised, eigendecomposed, and floored:

$$A = Q \text{diag}(d_1, \dots, d_p) Q^T, \quad \tilde{A} = Q \text{diag}(\max(d_i, \varepsilon s)) Q^T, \quad (45)$$

where $s = \max(\max_i |d_i|, 1)$ and $\varepsilon = 10^{-7}$. The number of clipped eigenvalues and before/after minima are returned for audit. In the empirical core diagnostics, no factor or residual covariance eigenvalues required clipping. Whitening separately uses a numerical floor of 10^{-12} .

9 Backtest accounting and performance measures

9.1 Holdings drift and turnover

If target weight $w_{i,0}$ is held through asset return $r_{i,t}$, the post-return weight is

$$w_{i,t}^- = \frac{w_{i,t-1}(1 + r_{i,t})}{1 + w_{t-1}^\top r_t}. \quad (46)$$

At rebalance j , reported one-way turnover is

$$\text{TO}_j = \frac{1}{2} \|w_j - w_j^-\|_1. \quad (47)$$

The code charges $c\|w_j - w_j^-\|_1 = 2c \text{TO}_j$ on the first holding day, with $c = 10$ basis points for each unit of gross bought-or-sold notional. Inception turnover is set to zero, so initial establishment cost is omitted. Annualised turnover is total one-way turnover divided by out-of-sample years.

9.2 Return and risk definitions

For daily returns $x_1, \dots, x_n$ and $A = 252$:

$$\text{annualised geometric return} = \left[\prod_{t=1}^n (1 + x_t) \right]^{A/n} - 1, \quad (48)$$

$$\text{annualised volatility} = \sqrt{A} \text{sd}(x_t), \quad (49)$$

$$\text{Sharpe-like ratio} = \sqrt{A} \frac{\bar{x}}{\text{sd}(x_t)}, \quad (50)$$

$$\text{maximum drawdown} = \min_t \left(\frac{W_t}{\max_{s \leq t} W_s} - 1 \right), \quad (51)$$

$$\text{reported daily ES} = \sqrt{A} \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} (-x_t), \quad (52)$$

where $W_t = \prod_{s \leq t} (1 + x_s)$ and $\mathcal{T}$ contains losses at or above the empirical 95th percentile. Sharpe uses gross returns in the main performance table, while “net return” uses returns after transaction costs.

Important limitation. Multiplying daily empirical ES by $\sqrt{252}$ is a reporting convention, not an exact temporal aggregation law. It is exact only under restrictive scale-stable assumptions. The formal comparison therefore also works directly with non-overlapping monthly portfolio losses.

10 Statistical analysis

10.1 Monthly paired moving-block bootstrap

Daily net portfolio returns are compounded within calendar month:

$$R_{m,k} = \prod_{t \in m} (1 + r_{t,k}^{\text{net}}) - 1, \quad L_{m,k} = -R_{m,k}. \quad (53)$$

For strategies a and b , the statistics are differences in annualised monthly-loss volatility, monthly ES, and annualised geometric net return:

$$\Delta_\sigma = \sqrt{12} [\text{sd}(L_a) - \text{sd}(L_b)], \quad (54)$$

$$\Delta_{ES} = \text{ES}_{0.95}(L_a) - \text{ES}_{0.95}(L_b), \quad (55)$$

$$\Delta_R = R_a^{\text{ann}} - R_b^{\text{ann}}. \quad (56)$$

Each bootstrap draw samples circular blocks of three consecutive months until the original number of months is reached. The same month indices are applied to every strategy, preserving paired cross-strategy shocks and short-run serial dependence [12]. The project uses 1,999 draws and percentile 95% intervals.

The reported two-sided bootstrap probability is

$$p^* = 2 \min\{\Pr(\Delta^* \leq 0), \Pr(\Delta^* \geq 0)\}, \quad (57)$$

capped at one. It is shown only for pre-specified pairs. This is a sign-probability from the uncentred bootstrap distribution, not a studentised null-recentred test. It is useful descriptive evidence, but a journal submission should add a formally null-imposed block-bootstrap or studentised procedure.

Core primary comparisons were B versus C and C versus D. Matched primary comparisons were matched B versus D and matched C versus D. Improved-run primary comparisons were empirical D versus D-FS and matched B versus D-REG. All other pairwise intervals are exploratory and deliberately have missing p-values. No family-wise or false-discovery adjustment is applied; robustness sweeps are not treated as hypothesis tests.

10.2 Ex-ante volatility regimes

Regimes use the NIFTY index, never portfolio outcomes. Let

$$v_t = \sqrt{252} \text{sd}(r_{t-20:t}^{\text{index}}). \quad (58)$$

The signal is shifted one day. The threshold is the expanding historical 75th percentile of the shifted signal, requiring at least 126 observations and itself shifted once more. Day t is high volatility when yesterday's known v exceeds the threshold known before day t . Missing early classifications default to normal.

The core interaction asks whether C's risk benefit over B is larger in high volatility:

$$[M(B, H) - M(C, H)] - [M(B, N) - M(C, N)], \quad (59)$$

for volatility and ES. The improved interaction analogously compares regularised CVaR with empirical CVaR and adds arithmetic annualised net return. Circular daily blocks have length 21 and use 1,999 draws.

10.3 Covariance forecast quality

For rebalance j , the realised proxy is the sample covariance of daily asset returns over the next monthly holding interval, $\Sigma_{j+1}^{\text{real}}$. Two losses are

$$\mathcal{L}_F = \|\hat{\Sigma}_j - \Sigma_{j+1}^{\text{real}}\|_F, \quad (60)$$

$$\mathcal{L}_w = \left| w_j^T \Sigma_{j+1}^{\text{real}} w_j - w_j^T \hat{\Sigma}_j w_j \right|. \quad (61)$$

The first evaluates the whole matrix; the second evaluates forecast error along the selected portfolio direction. The realised proxy is noisy because each month contains only about 20 observations for 39 assets. In simulation, the future true covariance is known and averaged across the 63-day evaluation interval.

10.4 Factor diagnostics

At each rebalance the project records:

1. retained variance share: sum of the first K weighted singular values squared divided by the sum over all singular values;
2. loading-space stability: the largest principal angle between consecutive endpoint loading column spaces;
3. residual-correlation sparsity: fraction of off-diagonal thresholded correlations equal to zero;
4. Fisher excess kurtosis of each factor score series, averaged across factors;
5. median Fisher excess kurtosis across residual series;
6. eigenvalue clipping counts and EWMA effective sample size.

The final-window standardised factors and residuals receive normal QQ plots. No large-sample omnibus normality-test p-values are used because tiny deviations become mechanically significant at large sample sizes.

11 Empirical results

11.1 Core full-sample comparison

Table 4: Core NIFTY results, February 2017–August 2026

Model	Gross return	Volatility	Sharpe	Max DD	ES 95%	Turnover	Net return	Net wealth
Equal weight	19.81%	15.94%	1.215	-35.79%	36.86%	0.31	19.74%	5.411
Original TV-MVP	15.88%	14.37%	1.098	-30.04%	31.81%	6.63	14.35%	3.515
Dynamic- Σ_F TV-MVP	15.74%	14.36%	1.091	-30.34%	31.81%	6.63	14.22%	3.477
Dynamic-factor CVaR	13.44%	14.19%	0.960	-34.37%	31.65%	4.73	12.37%	2.983

The dynamic factor covariance is nearly indistinguishable from the original. CVaR has slightly lower daily volatility and ES than the unconstrained original, but this comparison mixes objective and feasible-set effects. Equal weight’s large drawdown shows that its high return and Sharpe did not imply lower downside magnitude.

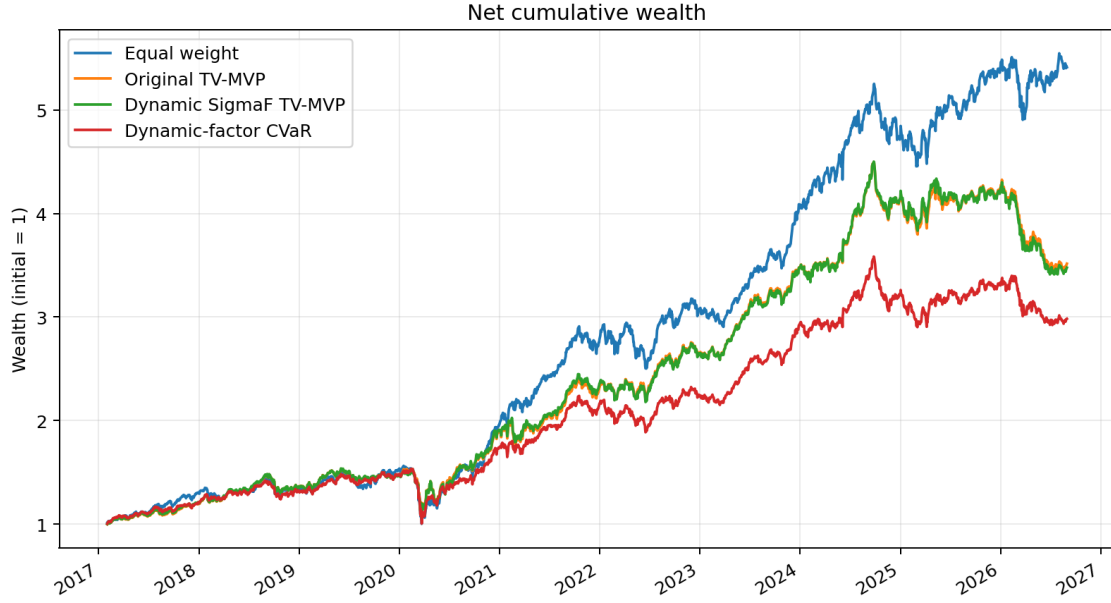


Figure 2: Full-sample cumulative net wealth for the core comparison.

11.2 Primary monthly inference

Table 5: Pre-specified core comparisons using monthly losses

Pair (1 minus 2)	Metric	Estimate	CI low	CI high	p^*
B minus C	Ann. volatility	0.00038	-0.00198	0.00238	0.755
B minus C	Monthly ES	0.00198	-0.00580	0.00901	0.434
B minus C	Ann. net return	0.00131	-0.00766	0.01089	0.781
C minus D	Ann. volatility	-0.00701	-0.02382	0.01072	0.498
C minus D	Monthly ES	-0.01492	-0.03600	0.00183	0.119
C minus D	Ann. net return	0.01804	-0.01883	0.05471	0.358

All intervals contain zero. Nothing in the full-sample primary inference supports a statistically distinguishable advantage for dynamic Σ_F or baseline CVaR.

11.3 Regime results

High-volatility days number 560; normal-volatility days number 1,802. B and C have virtually identical high-regime volatility (22.30%) and ES (53.16% versus 53.13%). D has 21.75% volatility and 50.72% ES in the high regime, but lower normal-regime return and Sharpe. The high-minus-normal dynamic-covariance benefit is -0.00018 for volatility (95% CI -0.00404, 0.00396; $p^* = 0.906$) and 0.00103 for ES (CI -0.01585, 0.01734; $p^* = 0.975$). There is no evidence that extension C becomes more useful when index volatility is high.

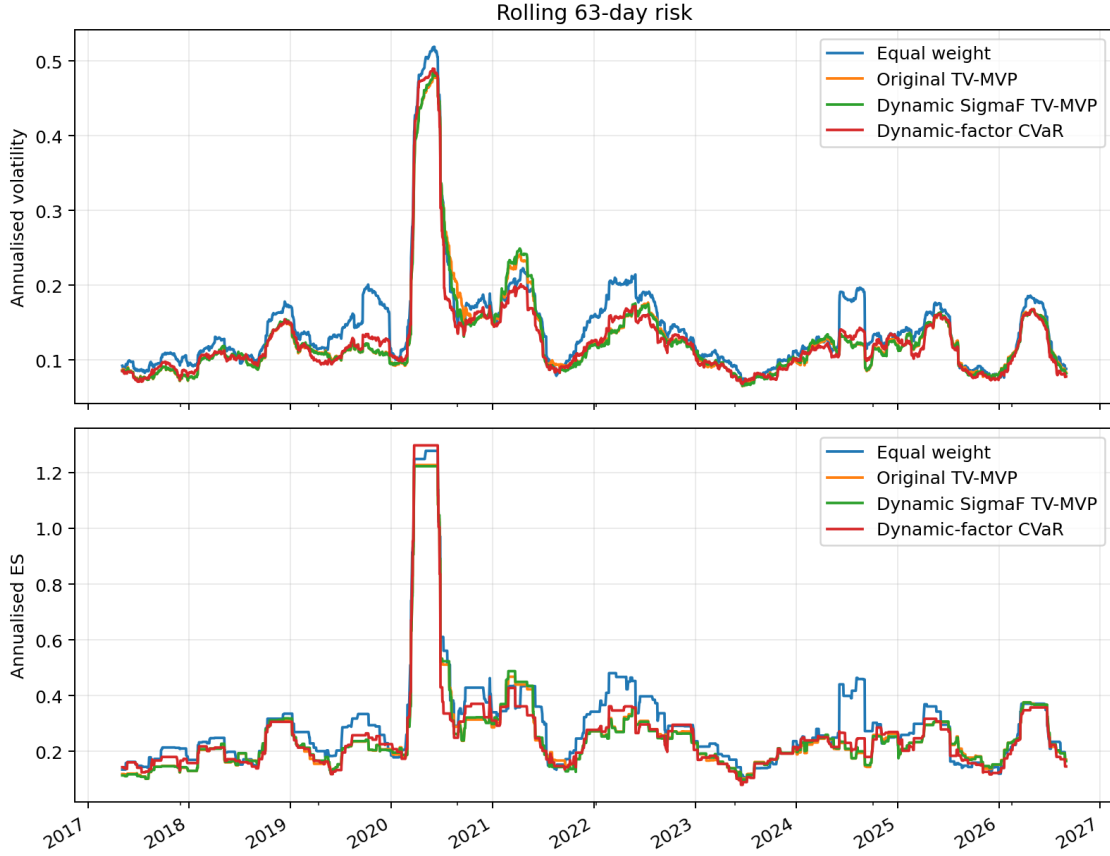


Figure 3: Rolling 63-day annualised volatility and empirical ES. The common temporal movement dominates the modest cross-strategy differences.

11.4 Covariance forecast evaluation

Table 6: Mean next-holding-period covariance forecast losses

Model	Frobenius loss	Portfolio-risk loss
Original TV-MVP	0.005623	0.000057
Dynamic- Σ_F TV-MVP	0.005958	0.000058
CVaR scenario covariance	0.005573	0.000052

Dynamic factor covariance does not improve either mean loss. The CVaR scenario covariance has slightly smaller averages, yet its portfolio is not better after matching constraints. A globally good covariance approximation does not guarantee accurate estimation of a small tail functional, and portfolio optimisation can amplify local forecast errors.

11.5 Factor-model diagnostics

Three factors explain 48.91% of locally weighted variation on average, ranging from 35.38% to 76.78%. Consecutive loading subspaces have a mean maximum principal angle of 54.48 degrees and a maximum of 89.98 degrees, indicating substantial endpoint instability. Only 14.47% of off-diagonal residual correlations are exactly thresholded to zero. Mean factor excess kurtosis is 2.43; median residual excess kurtosis is 2.49. These values and the QQ plots demonstrate much

heavier tails than Gaussian benchmarks.

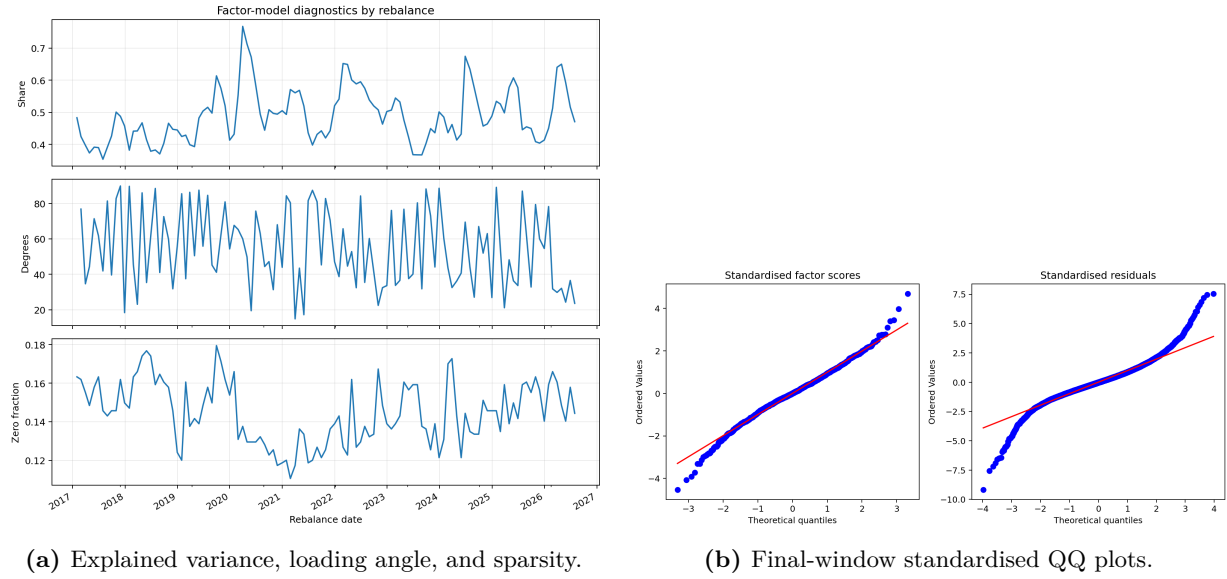


Figure 4: Factor diagnostics. Large principal angles and heavy-tailed scores/residuals are directly relevant to covariance and CVaR instability.

12 Why empirical CVaR failed after 2021

12.1 Removing the feasible-set confound

The original post-2021 comparison was partly confounded: B was unconstrained whereas D was long-only and capped. The matched rerun applies Equation (23) to both MVP estimators and CVaR.

Table 7: Matched post-2021 comparison

Model	Gross return	Net return	Volatility	Sharpe	Max DD	ES 95%	Turnover
Equal weight	20.14%	20.07%	13.87%	1.393	-15.15%	32.09%	0.29
Original-covariance MVP	13.62%	12.79%	11.86%	1.136	-17.13%	26.17%	3.69
Dynamic-covariance MVP	13.52%	12.62%	11.92%	1.123	-16.63%	26.25%	3.99
Empirical CVaR	11.22%	10.17%	12.13%	0.937	-18.68%	27.23%	4.75

Under the same constraints, empirical CVaR is worse than original MVP on every displayed measure. The monthly block-bootstrap comparison B minus D gives annualised volatility difference -0.00862 (CI -0.01522, -0.00219; $p^* = 0.010$), monthly ES difference -0.01061 (CI -0.01944, -0.00139; $p^* = 0.031$), and annualised net return difference 0.02565 (CI 0.00083, 0.05109; $p^* = 0.043$). Within this sample and test definition, matched B is distinguishable at 5% on all three registered measures.

12.2 Forecast-tail reversal

Empirical CVaR succeeds inside its fitted scenario distribution: mean fitted daily CVaR is 1.106%, below matched original MVP's 1.226%. Out of sample, the ordering reverses: realised holding-period daily CVaR averages 1.326% for D and 1.295% for B. D's fitted-to-realised gap is 0.219

percentage point, its VaR exceedance rate is 10.25%, and its forecast/realised CVaR rank correlation is only 0.314.

This pattern distinguishes three explanations:

1. **Not a solver bug:** the LP finds a portfolio with lower objective value in the supplied scenarios.
2. **Not merely the cap:** the reversal remains when MVP has the same cap and long-only rules.
3. **Scenario/optimisation interaction:** a small, under-responsive empirical tail is treated as known, and the optimiser magnifies its sampling error.

12.3 Corner solutions and costs

Baseline CVaR holds 12.52 effective assets on average, places 4.91 weights at the cap, and sets 22.51 of 39 weights to zero. Matched original MVP holds 14.23 effective assets, with 3.18 cap hits and 18.99 zeros. D’s annual turnover is 4.75 versus B’s 3.69. The gross return gap already exists, so costs are not the sole cause; costs widen it.

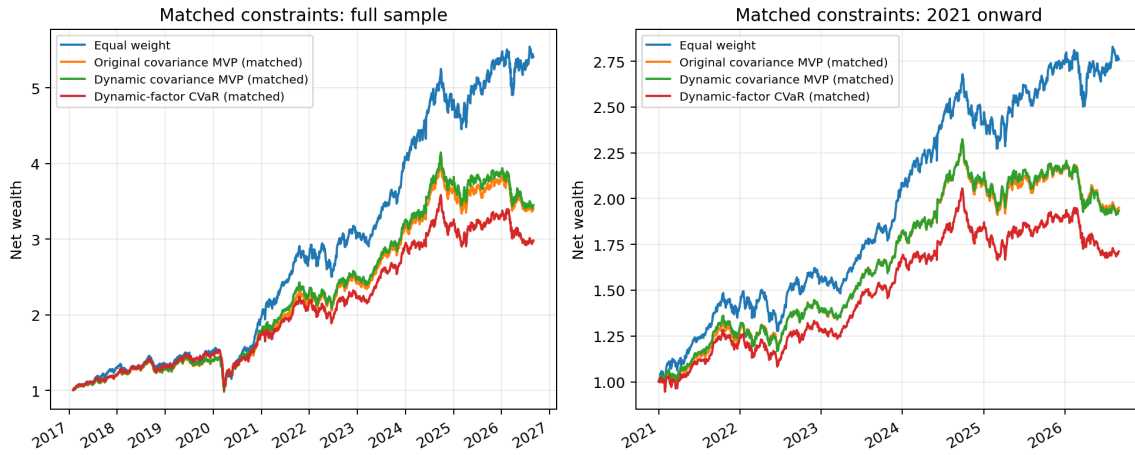


Figure 5: Cumulative net wealth for the matched-constraint experiment, over the full period and after 2021.

13 Improved CVaR results

13.1 Ablation results

Table 8: Post-2021 improved-CVaR ablation; all optimisers matched

Model	Gross return	Net return	Volatility	Sharpe	Max DD	ES 95%	Turnover
Original-covariance MVP	13.62%	12.79%	11.86%	1.136	-17.13%	26.17%	3.69
Empirical CVaR	11.22%	10.17%	12.13%	0.937	-18.68%	27.23%	4.75
Filtered/stressed CVaR	13.92%	12.88%	12.10%	1.138	-16.94%	27.00%	4.60
Regularised filtered CVaR	15.10%	14.30%	11.95%	1.237	-15.67%	26.56%	3.50

Filtering and stressing add 2.71 percentage points of annualised net return relative to empirical CVaR, with nearly unchanged daily volatility and a 0.23-point reduction in annualised daily ES. Adding regularisation contributes another 1.42 points of net return, reduces volatility by 0.15

point, improves maximum drawdown by 1.27 points, reduces ES by 0.44 point, and removes 1.10 units of annual turnover.

Relative to matched original MVP, D-REG has 1.51 points higher net return, a 0.100 higher Sharpe ratio, and a 1.46-point shallower maximum drawdown. It also has 0.09 point higher daily volatility and 0.39 point higher annualised daily ES. Thus even the point-estimate verdict depends on the performance measure.

13.2 Calibration after the repair

Table 9: Post-2021 tail calibration and concentration

Model	Scenario CVaR	Realised CVaR	Gap	Exceedance	Rank corr.	Effective assets
Empirical CVaR	0.01106	0.01326	0.00219	10.25%	0.314	12.52
Filtered/stressed CVaR	0.01286	0.01309	0.00023	8.63%	0.446	12.06
Regularised filtered CVaR	0.01282	0.01287	0.00005	8.14%	0.446	12.56
Matched original MVP [†]	0.01431	0.01295	-0.00136	7.13%	0.451	14.23

[†]For the improved diagnostic, non-empirical-CVaR models are evaluated on the same filtered/stressed scenario system. Scenario CVaR is a forecast diagnostic for MVP, not its optimisation objective.

The stress mixture removes most average underprediction and increases rank correlation, but exceedance remains well above the nominal 5%. The filtered optimiser alone becomes slightly more concentrated; most of the trading improvement comes from explicit regularisation, not from filtering automatically creating diversified weights.

13.3 Statistical uncertainty

For the pre-specified empirical-D minus D-FS comparison, annualised monthly volatility difference is 0.00470 (CI -0.00458, 0.01425; $p^* = 0.315$), monthly ES difference is 0.01408 (CI -0.00080, 0.02552; $p^* = 0.071$), and annualised net-return difference is -0.02650 (CI -0.05916, 0.00365; $p^* = 0.097$). These are suggestive but not 5%-significant.

For matched B minus D-REG, annualised monthly volatility difference is -0.00451 (CI -0.01182, 0.00255; $p^* = 0.221$), monthly ES difference is 0.00258 (CI -0.00513, 0.00902; $p^* = 0.497$), and annualised net-return difference is -0.01473 (CI -0.04282, 0.01151; $p^* = 0.275$). None is statistically distinguishable from zero.

Exploratory empirical-D minus D-REG intervals exclude zero for monthly ES and annualised net return, but that pair was not designated primary and receives no p-value. This protects the analysis from relabelling the most favourable post-hoc pair as confirmatory.

13.4 High- versus normal-volatility performance

After 2021, 313 days are classified high volatility and 1,086 normal. In high volatility, empirical D versus D-REG has volatility 16.05% versus 16.13%, and ES 33.69% versus 32.83%. In normal volatility, the corresponding figures are 10.71% versus 10.41%, and 24.91% versus 23.76%. The 21-day-block interaction test does not show a larger high-regime benefit: volatility interaction -0.00377 ($p^* = 0.234$), ES interaction -0.00292 ($p^* = 0.740$), and arithmetic net-return interaction 0.01547 ($p^* = 0.665$).

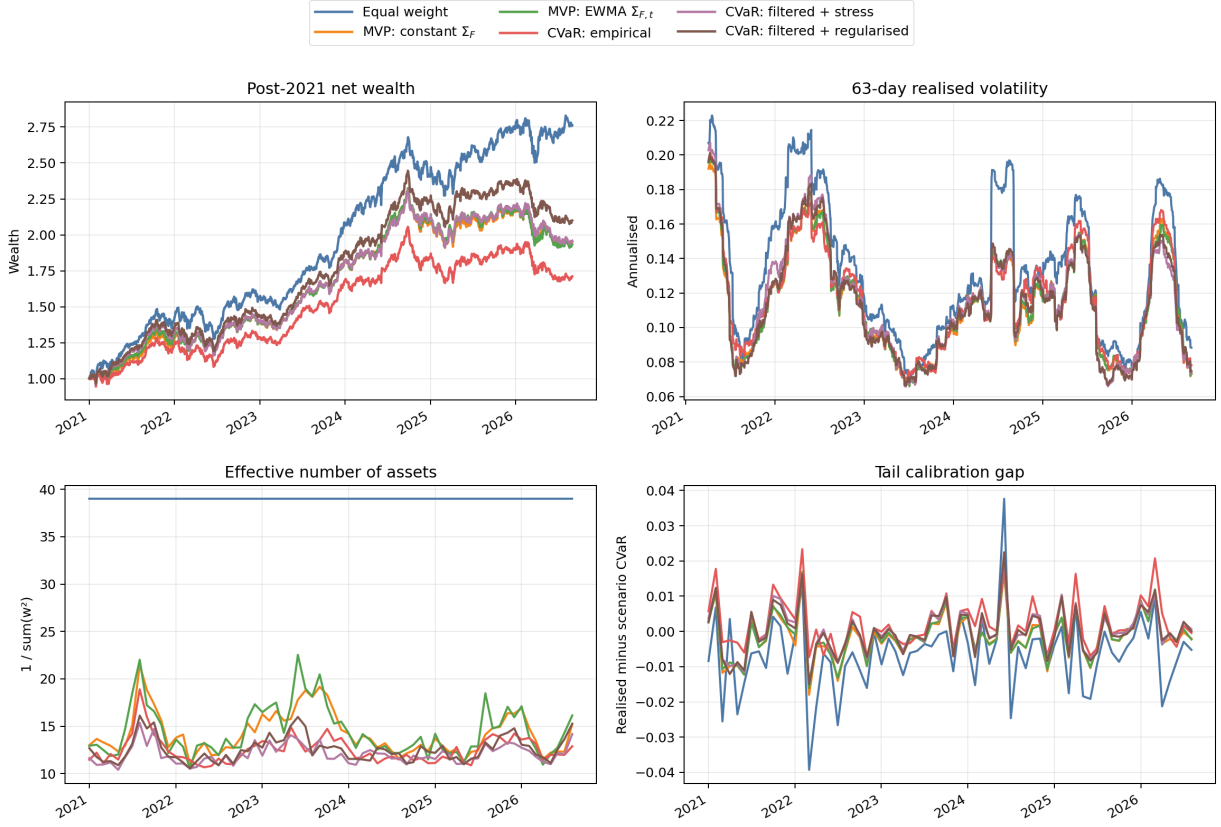


Figure 6: Post-2021 improved-CVaR diagnostics. The shared legend identifies all six series. Regularised filtered CVaR improves wealth and turnover-related stability, but risk paths remain close and calibration gaps still spike in some months.

14 Simulation evidence

14.1 Design

Each of three settings has 100 independent replications, with fixed but changeable project seed, $p = 20$, $K = 2$, 180 estimation days, and 63 evaluation days. Loadings evolve smoothly:

$$B_t = B_0 + D \sin(2\pi t / (T + H - 1)). \quad (62)$$

Base factor covariance is a scaled 2×2 matrix with off-diagonal correlation structure, and residual covariance is Toeplitz. The factor scale also rises gradually with time.

Gaussian: factors and residuals are Gaussian.

Student- t : both use multivariate scale mixtures with five degrees of freedom, rescaled so covariance matches the designated covariance.

Regime shift: beginning 20 days before the estimation endpoint, factor variance is multiplied by four, residual variance by 2.25, and residual correlation parameter rises from 0.20 to 0.65. The estimator therefore sees only 20 shifted training observations before forecasting 63 shifted days.

The future true covariance is known and averaged across evaluation days. For every metric the summary reports replication mean, standard deviation, and normal-approximation Monte Carlo interval $\bar{x} \pm 1.96s/\sqrt{100}$.

Important limitation. The simulation’s original/dynamic MVPs are unconstrained, whereas simulated CVaR is long-only and capped. Those probability comparisons remain feasible-set confounded. The improved filtered/regularised CVaR is not included in the current simulation module. A research-paper extension should rerun simulations with matched constraints and the full ablation.

14.2 Results

Table 10: Probability of beating original TV-MVP over 100 replications

Setting	Method	Realised variance	Realised ES
Gaussian	Equal weight	0.67	0.65
Gaussian	Dynamic Σ_F	0.48	0.47
Gaussian	Empirical CVaR	0.27	0.31
Student- t	Equal weight	0.54	0.53
Student- t	Dynamic Σ_F	0.54	0.60
Student- t	Empirical CVaR	0.22	0.29
Regime shift	Equal weight	0.71	0.64
Regime shift	Dynamic Σ_F	0.21	0.19
Regime shift	Empirical CVaR	0.64	0.62

Dynamic Σ_F does not reliably beat original TV-MVP. With $\lambda = 0.94$, its effective sample is short and its covariance estimate is noisy; in the abrupt shift design it does especially poorly. Baseline CVaR is more competitive in the regime shift, consistent with tail scenarios reacting to recently observed shocks, but fails to dominate Gaussian or Student- t settings. Equal weight is surprisingly competitive, emphasising estimation-error costs.

Mean true-covariance Frobenius errors for original, dynamic, and scenario covariance are respectively 0.000605, 0.000638, and 0.000622 under Gaussian returns; 0.000845, 0.000888, and 0.000851 under Student- t ; and 0.001943, 0.002110, and 0.001964 under regime shift. Dynamic EWMA does not improve true covariance error in any design.

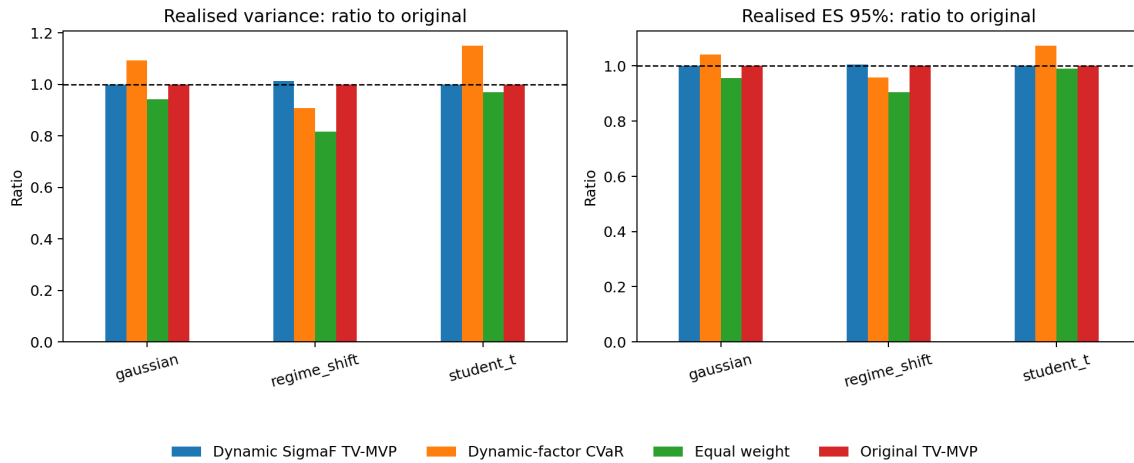


Figure 7: Mean realised variance and ES ratios to original TV-MVP. Values below one favour the displayed method, subject to the constraint caveat.

15 Robustness and sensitivity

The robustness grid varies one parameter at a time from the common baseline and evaluates 1,399 post-2021 observations:

- EWMA decay: 0.90, 0.94, 0.97;
- rolling window: 378, 504, 756 trading days;
- factor number: 2, 3, 5;
- CVaR confidence: 0.90, 0.95, 0.99;
- transaction cost: 0, 10, 25 basis points.

C-minus-B Sharpe difference remains negative across every EWMA decay (-0.0339 to -0.0187), window (-0.0461 to -0.0209), and factor count (-0.0422 to -0.0290). C-minus-B net return is also negative throughout. Baseline D-minus-B Sharpe varies substantially with confidence: -0.0154 at 90%, -0.1332 at 95%, and -0.0049 at 99%. D-minus-B ES changes sign at 99%. This instability is a sensitivity failure, not evidence for selecting 99% after seeing the outcome.

Transaction costs monotonically worsen the dynamic model’s net-return difference because turnover is high. The transaction-cost sweep recomputes net returns from fixed gross returns and turnover; it does not refit weights.

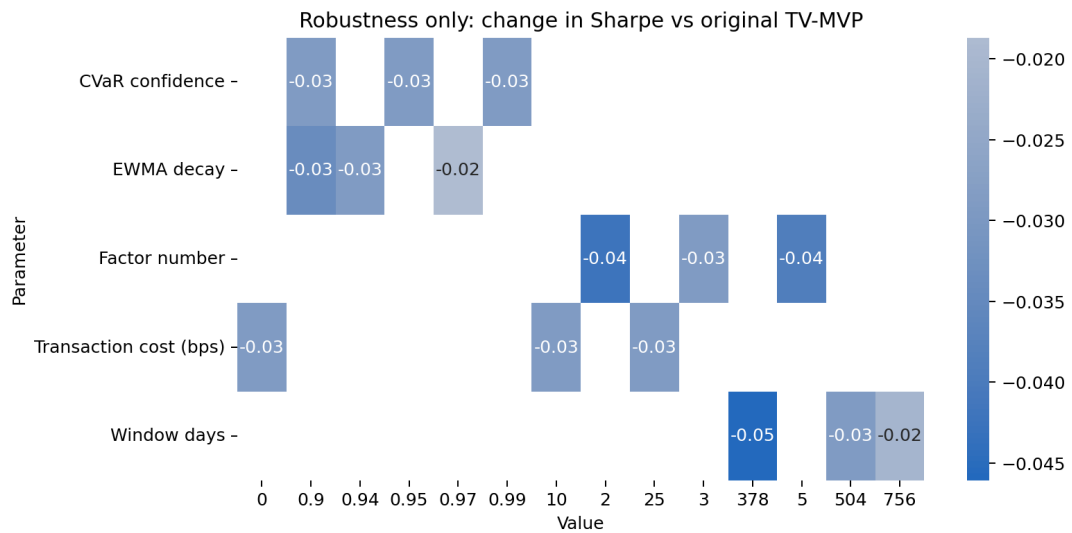


Figure 8: Compact robustness heatmap. It shows dynamic-minus-original Sharpe only; CVaR sensitivity remains available in the CSV table.

16 Software implementation in detail

16.1 Module-to-method map

Table 11: Python implementation map

File	Responsibility
<code>config.py</code>	YAML loading, required-section validation, and deep-copy dotted overrides.

File	Responsibility
<code>data.py</code>	Author MAT loading; Yahoo cache/download; completeness filter; adjusted-close returns; monthly endpoint positions; SHA-256 data hashes.
<code>factor.py</code>	Rule-of-thumb bandwidth, discrete Epanechnikov weights, local SVD/PCA at every date, least-squares scores and residuals, factor count approximation, principal angles.
<code>covariance.py</code>	Sparse residual correlation thresholding, ordinary and EWMA factor covariance, positive-definite eigenvalue repair.
<code>optimisation.py</code>	Equal weights, unconstrained/constrained MVP, empirical CVaR LP, filtered scenario generator, and regularised CVaR LP.
<code>backtest.py</code>	Shared rolling engine, matched/improved strategy modes, holdings drift, costs, covariance losses, diagnostics, scenario calibration, and ex-ante regimes.
<code>metrics.py</code>	Geometric return, volatility, Sharpe, maximum drawdown, empirical ES, monthly compounding, rolling ES, regime tables.
<code>inference.py</code>	Circular moving-block indices, all-pair intervals, primary-only p-values, original dynamic-regime test, and improved CVaR regime interaction.
<code>reproduction.py</code>	NumPy author DGP, supplied example workflow, reproduction tables and plots.
<code>empirical.py</code>	Core empirical orchestration, output writing, plots, and one-at-a-time sensitivity.
<code>fair_comparison.py</code>	Constraint-matched performance, monthly inference, calibration, yearly results, and cumulative plots.
<code>cvar_diagnostics.py</code>	Post-2021 mechanism analysis: calibration, concentration, allocation, factor shift, yearly performance, exploratory block intervals.
<code>improved_cvar.py</code>	Filtered/stressed/regularised ablation, primary inference, regime interaction, calibration, concentration, and corrected shared-legend plot.
<code>simulation.py</code>	Three DGPs, parallel replication, known-covariance error, Monte Carlo intervals, beat probabilities.
<code>plotting.py</code>	Core cumulative, rolling-risk, weights/turnover, factor, QQ, and sensitivity graphics.
<code>manifest.py</code>	Run time, platform, Git commit, seed, config hash, and package versions.

16.2 Numerical solvers

Unconstrained MVP uses `numpy.linalg.solve`; matched MVP uses SciPy SLSQP with analytic gradient $2\Sigma w$, equality constraint, and box bounds. CVaR and regularised CVaR use `scipy.optimize.linprog(method="highs")` with explicit slack variables. All successful targets are checked indirectly in tests and output audits. In the final improved output, weight sums deviate from one by at most 2.21×10^{-11} , minimum weight is zero, and maximum weight is exactly 0.10.

16.3 Randomness and reproducibility

Project seed is 20260906. At empirical rebalance number j , baseline scenarios use seed $20260906 + j$ and improved scenarios use $20260906 + 50000 + j$. Bootstrap inference uses separately offset seeds. Simulations use offsets by DGP and replication. The manifest records Python 3.11.0 and package versions including NumPy 2.2.6, pandas 2.3.1, SciPy 1.16.0, Matplotlib 3.10.3, seaborn 0.13.2, PyYAML 6.0.3, and yfinance 0.2.65.

16.4 Tests

Seven tests currently pass:

1. indefinite covariance matrices are repaired to positive definite;
2. EWMA reacts to recent high variance;
3. MVP and CVaR weights satisfy full-investment and cap constraints;
4. filtered scenarios reproduce exactly under an identical seed, and regularised weights satisfy constraints;
5. circular block indices are reproducible and contiguous within blocks;
6. the regime classifier does not use the current return shock;
7. changing future returns does not change earlier weights; matched optimisers also satisfy bounds and sums.

The tests do not prove statistical correctness. Missing tests include equivalence to author MATLAB outputs under the same RNG, formal KKT residual checks for every optimisation, data-vendor revision detection, and end-to-end snapshot tests for every table.

17 Limitations, threats to validity, and honest interpretation

17.1 Data validity

- Current constituents create survivorship bias and potentially favour stable winners.
- Yahoo adjusted-close histories can contain vendor adjustments and are not an institutional-quality point-in-time database.
- The sample covers one national equity market and one broad post-2015 period.
- Returns are not excess returns; the Sharpe-like measure assumes zero risk-free rate.
- Volume, spreads, borrow availability, corporate-action audit trails, taxes, and market impact are absent.

17.2 Model validity

- The factor number is fixed at three empirically; factor-number uncertainty is not propagated.
- Local loading spaces can rotate sharply, and sign alignment cannot resolve general rotations within nearly tied eigenspaces.
- Residual covariance is treated as constant in B/C even though diagnostics motivate a dynamic alternative.
- The portable soft-threshold estimator is not the authors' `spcov` estimator.
- EWMA decay 0.94 implies only 32.33 effective observations and can be noisy.
- Empirical and filtered scenarios condition on estimated latent quantities without accounting for parameter uncertainty.
- Stress probability and multipliers are researcher-chosen rather than estimated from an independent training sample.

17.3 Evaluation validity

- A one-month realised covariance is rank deficient when $p = 39$ and contains substantial sampling noise.
- Daily ES annualisation by square-root-of-time is heuristic.
- Monthly ES is estimated from roughly 115 months in the full sample and only 68 post-2021 months; the 5% tail is therefore extremely small.
- Moving blocks address short serial dependence but block length three is itself a modelling choice.
- The bootstrap p-value is an uncentred sign probability, not a null-imposed studentised test.
- Overlapping 504-day estimation windows make successive portfolios highly dependent even when realised monthly returns do not overlap.
- Many robustness configurations were examined. They are labelled exploratory, but researcher degrees of freedom remain.
- The improved model was motivated by observed baseline failure. Its existing NIFTY performance is not a clean holdout validation.

17.4 Economic interpretation

Equal weight has the strongest realised return and Sharpe. This may reflect a broad equal-weight premium, omitted estimation error, or survivorship-biased constituent selection; the project cannot identify causality. MVP portfolios achieve lower risk but sacrifice return. Raw CVaR is not inherently defective: it fails because its estimated tail distribution is a poor enough guide to the next month that optimisation overfits it. The filtered/regularised method improves the forecast inputs and controls the optimiser's freedom, but its apparent return advantage over matched MVP is statistically uncertain.

Backtest significance would not prove future profitability. Even a correctly sized historical test conditions on the market, assets, rules, costs, and structural regime that generated the sample. Deployment introduces model drift, liquidity feedback, strategy crowding, and unobserved implementation costs.

18 Research programme for a publishable extension

The repository supports a credible next-stage question:

Does conditionally scaled and regularised factor-based CVaR improve tail-risk portfolio selection relative to time-varying minimum variance, and under which data-generating regimes?

A defensible confirmatory protocol should do the following before viewing new outcomes:

1. Freeze D-FS and D-REG Version 1, including every multiplier and penalty.
2. Designate primary estimands: realised monthly ES, tail-calibration score, turnover, and net return; choose a small number of directional hypotheses.
3. Obtain point-in-time constituents and delisting returns for NIFTY and at least two independent markets.
4. Use nested walk-forward tuning: at rebalance t , choose decay, stress, and regularisation only with data dated before t .
5. Compare all strategies under both common matched constraints and a separate unconstrained/realistic mandate analysis.

6. Add formal VaR coverage and independence diagnostics, ES forecast scoring, and conditional calibration tests.
7. Add covariance forecast tests that respect noisy realised proxies, possibly with high-frequency realised covariance where available.
8. Expand simulation axes: tail index, factor-strength, residual sparsity, loading smoothness, covariance break size, break timing, p/T , and transaction costs.
9. Include the full improved-CVaR ablation in at least 100, preferably 500–1,000, matched-constraint replications per cell.
10. Use a null-recentred or studentised dependent bootstrap and state a multiple-testing adjustment for the primary family.
11. Preserve a genuinely untouched market or future period for final evaluation.

The strongest defensible contribution is not “CVaR beats MVP.” It is that raw empirical CVaR is unusually sensitive to conditional-distribution and small-tail-sample error, while current-state filtering and portfolio regularisation can reduce calibration bias and optimiser instability. Whether this translates into robust economic superiority remains an open empirical question.

19 Reproduction instructions

From the repository root, using Python 3.10 or later:

```
python3 -m pip install -e .
python3 scripts/run_all.py
```

Focused commands are:

```
python3 scripts/run_reproduction.py
python3 scripts/run_empirical.py --with-sensitivity
python3 scripts/run_cvar_diagnostics.py
python3 scripts/run_fair_comparison.py
python3 scripts/run_improved_cvar.py
python3 scripts/run_simulations.py
```

The frozen data are used unless `-force-download` is explicitly supplied to a compatible entry point. Changing `project.seed` in `config.yaml` changes stochastic draws. The all-in-one run updates `results/run_manifest.json`. The improved empirical backtest is computationally heavier because each rebalance solves multiple LPs and performs factor/residual whitening.

A Algorithmic pseudocode

A.1 Common rolling backtest

```
for each completed month-end t after 504 observations:
    estimation = returns[t-503 : t]
    holding    = returns[t+1 : next_month_end]

    fit local PCA at every date in estimation
    estimate score path and residual-vector path
    estimate sparse residual covariance
    estimate sample and EWMA factor covariance
    construct original and dynamic return covariance
    generate paired block-bootstrap scenarios
```

```

solve targets for A, B, C, and D
if matched/improved mode:
    impose identical long-only 10% cap on optimisers
if improved mode:
    generate filtered/stressed scenarios
    solve filtered and regularised CVaR targets

compute turnover from drifted previous holdings
charge trading cost on first holding day
realise portfolio using daily drifted weights
evaluate forecast covariance against next-month proxy
record factor, tail-calibration, and concentration diagnostics

```

A.2 Moving-block inference

```

compound each strategy's net daily returns to calendar months
compute observed statistic for each strategy pair
for b = 1,...,1999:
    draw circular three-month blocks until sample length is reached
    apply identical month indices to both strategies
    recompute volatility, ES, and annualised net-return differences
report percentile 2.5% and 97.5% quantiles
report sign-probability p* only for registered pairs

```

B Complete configuration snapshot

Table 12: Key configuration values

Parameter	Value
Seed; workers; annualisation	20260906; 4; 252
Data dates	2015-01-01 to 2026-09-05 exclusive
Index	̂NSEI
Minimum history fraction	0.985
Rolling window; rebalance	504 trading days; month-end
Factors; max reproduction factors	3; 6
Bandwidth	Rule of thumb, Equation (9)
Residual threshold scale	0.15
Eigenvalue floor	10^{-7} times covariance scale
Factor EWMA decay	0.94
Baseline CVaR	95%; 600 scenarios; 10-day blocks
Portfolio constraints	Long only; 10% cap for CVaR and matched runs
Improved scenarios	800; recent-start decay 0.995
Improved residual covariance	EWMA decay 0.97; dynamic blend weight 0.50
Stress mixture	15% probability; factor multiplier 1.50; residual multiplier 1.15
Regularisation	Turnover 0.0010; equal-weight distance 0.00025
Transaction costs	10 bp per unit gross traded notional

Parameter	Value
Post-COVID start	2021-01-01
Rolling plotted risk	63 days
High-volatility signal	21-day index volatility; expanding 75th percentile; minimum 126 days; lagged
Bootstrap	1,999 draws; three-month comparison blocks; 21-day regime blocks
Simulation	100 replications; 20 assets; 180 training; 63 evaluation; two factors; Student- t df 5; 350 scenarios
Sensitivity	$\lambda \in \{0.90, 0.94, 0.97\}$; $W \in \{378, 504, 756\}$; $K \in \{2, 3, 5\}$; CVaR $\in \{0.90, 0.95, 0.99\}$; costs $\in \{0, 10, 25\}$ bp

C Result provenance

Every major claim can be traced to a CSV rather than being read manually from a chart:

Evidence file	Content
<code>reproduction_results.csv</code>	Manual target comparison and portable simulation metrics
<code>empirical_performance.csv</code>	Full-sample A–D gross/net performance
<code>pairwise_bootstrap_inference.csv</code>	All core pair intervals and primary-only p-values
<code>regime_performance.csv</code>	High/normal metrics from ex-ante index regime
<code>regime_benefit_test.csv</code>	Dynamic-covariance high-regime interaction
<code>covariance_forecast_summary.csv</code>	Frobenius and portfolio-risk loss summaries
<code>factor_diagnostics.csv</code>	Rebalance-by-rebalance factor evidence
<code>simulation_replications.csv</code>	All 1,200 strategy-replication rows across three DGPs
<code>simulation_summary.csv</code>	Mean, standard deviation, Monte Carlo interval
<code>simulation_beat_probabilities.csv</code>	Probability each method beats original TV-MVP
<code>sensitivity_summary.csv</code>	One-at-a-time robustness grid
<code>matched_post_covid_performance.csv</code>	Fair post-2021 comparison
<code>matched_post_covid_inference.csv</code>	Fair monthly bootstrap evidence
<code>matched_tail_calibration.csv</code>	Fitted and realised tail behaviour, concentration
<code>improved_cvar_post_covid_performance.csv</code>	Improved-CVaR ablation performance
<code>improved_cvar_post_covid_inference.csv</code>	Improved primary and exploratory intervals
<code>improved_cvar_tail_calibration.csv</code>	Scenario method, bias, exceedance, ranking, weights
<code>improved_cvar_regime_interaction.csv</code>	Improved high-minus-normal bootstrap

D Glossary

Adjusted close Vendor price adjusted for corporate actions such as splits and distributions.

Block bootstrap Resampling consecutive observations rather than individual IID dates.

CVaR / expected shortfall Mean loss in the tail beyond a specified confidence quantile.

Effective assets Inverse Herfindahl concentration, $1/\sum_i w_i^2$.

EWMA Exponentially weighted moving average, assigning larger weight to recent data.

Factor loading Asset exposure to a latent common return shock.

Frobenius loss Square root of the sum of squared matrix-entry forecast errors.

Local PCA Principal components estimated with observations weighted by distance from a target date.

MVP Fully invested portfolio minimising variance for a supplied covariance matrix.

Principal angle Rotation-invariant measure of distance between factor-loading subspaces.

Scenario calibration gap Realised holding-period CVaR minus fitted scenario CVaR.

Survivorship bias Bias caused by constructing historical samples from securities known to survive to the present.

Turnover One-half the L_1 difference between target and drifted pre-trade weights.

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